

FOR THE MEETING · SUPERVISOR DISCUSSION SHEET

Paired PhD Programme — Points for Discussion

Candidates: Zina · Jamal Proposed supervisor: Prof. Reza Tarinejad October 2026

1. Proposed direction (subject to supervisor confirmation)

Programme	Dynamics of Flow-Excited Hydraulic Gates: Loading and Identification
Zina	Operational Modal Identification of Hydraulic Gates under Varying Flow Conditions. Identifying a gate's structural modes when water level and operation change its apparent dynamic properties (parameter-varying transmissibility OMA).
Jamal	Unsteady Hydrodynamic Loading of Sluice Gates with Bottom Sills. Regimes, mechanisms and dimensionless scaling of fluctuating gate loads, and an efficiency–dynamic-load criterion.
Basis	Connects Prof. Tarinejad's transmissibility-based identification research with the group's recent gate and weir hydraulics. Selected as the preferred paired direction after comparison with alternative architectures.

2. Five questions for Prof. Tarinejad

1. Could you advise whether Zina's proposed scope overlaps with the work of Farhad Amanzad or another current student?
2. Could you advise whether Jamal's proposed scope overlaps with the hydraulic-gate work of another current student?
3. Is a suitable hydraulic flume available or accessible, with tailwater control and the ability to hold still water at several levels?
4. Are the required pressure transducers, accelerometers, impact hammer and data-acquisition system available?
5. Would you prefer this integrated gate programme, or would you prefer one thesis to remain closer to the dam and seismic identification line?

3. To confirm before registration

Condition	Related question
Short recent-literature confirmation review of both gaps	Prepared by the candidates
No overlap with current students' topics	Questions 1 and 2
Flume and instrumentation available or accessible	Questions 3 and 4

Already reflected in the concepts: a bounded experimental scope (Zina about 20–30 runs; Jamal about 40–60 runs and 6–8 LES cases); Jamal's contribution framed at mechanism, scaling and criterion level; and clear scope separation with independent data ownership.

4. Scope separation (summary)

	Zina	Jamal
Specimen / sensors	Flexible gate; accelerometers; impact hammer	Rigid gate with sill inserts; pressure transducers
Modelling	Structural FE with added mass; synthetic loads	LES/DES (CFD); dimensional analysis
Validation	FE poles; EMA ground truth; public dam records	Measured spectra; held-out configurations
Shared	Flume, water supply, DAQ hardware, outline drawings	Common hydraulic-state labels (head, opening, submergence)
Independence	Remains executable if Jamal's work is delayed	Remains executable if Zina's work is delayed

5. If the conditions are not met

- **Laboratory not available:** both concepts become identification-based (Zina: uncertainty-quantified identification from seismic dam records; Jamal: gate identification, validated numerically and on an accessible regulator gate).
- **Zina's topic overlaps a current student's work:** Zina moves to the seismic-uncertainty concept; Jamal's concept is unchanged.
- **Jamal's topic overlaps a current student's work:** the same question is studied for labyrinth sliding gates with sills; otherwise the laboratory-independent alternative above applies.

6. Supervisor response

Paired direction	☐ Agree in principle	☐ Agree with changes	☐ Prefer alternative
Zina concept	☐ Proceed to proposal	☐ Modify	☐ Replace
Jamal concept	☐ Proceed to proposal	☐ Modify	☐ Replace
Notes and suggested changes			